

Review

Systematic review of rotavirus vaccination cost-effectiveness in high income settings utilising dynamic transmission modelling techniques

Tim Jesudason^a, Alejandra Rodarte^a, David Tordrup^a, Cristina Carias^b, Yao-Hsuan Chen^{c,*}^a Triangulate Health Ltd, United Kingdom^b MRL, Merck & Co., Inc., Rahway, NJ, USA^c Health Economic and Decision Sciences, MSD (UK) Limited, London, United Kingdom

ARTICLE INFO

Article history:

Received 31 May 2022

Received in revised form 16 June 2023

Accepted 16 June 2023

Available online 19 July 2023

Keywords:

Health economics

Rotavirus

Vaccination

Cost-effectiveness

ABSTRACT

Purpose: This systematic review presents cost-effectiveness studies of rotavirus vaccination in high-income settings based on dynamic transmission modelling to inform policy decisions about implementing rotavirus vaccination programmes.

Methods: We searched CEA Registry, MEDLINE, Embase, Health Technology Assessment Database, Scopus, and the National Health Service Economic Evaluation Database for studies published since 2002. Full economic evaluation studies based on dynamic transmission models, focusing on high-income countries, live oral rotavirus vaccine and children ≤ 5 years of age were eligible for inclusion. Included studies were appraised for quality and risk of bias using the Consensus on Health Economic Criteria (CHEC) list and the Philips checklist. The review protocol was prospectively registered with PROSPERO (CRD42020208406).

Results: A total of four economic evaluations were identified. Study settings included England and Wales, France, Norway, and the United States. All studies compared either pentavalent or monovalent rotavirus vaccines to no intervention. All studies were cost-utility analyses that reported incremental cost per quality-adjusted life year (QALY) gained. Included studies consistently concluded that rotavirus vaccination is cost-effective compared with no vaccination relative to the respective country's willingness to pay threshold when herd protection benefits are incorporated in the modelling framework.

Conclusions: Rotavirus vaccination was found to be cost-effective in all identified studies that used dynamic transmission models in high-income settings where child mortality rates due to rotavirus gastroenteritis are close to zero. Previous systematic reviews of economic evaluations considered mostly static models and had less conclusive findings than the current study. This review suggests that modelling choices influence cost-effectiveness results for rotavirus vaccination. Specifically, the review suggests that dynamic transmission models are more likely to account for the full impact of rotavirus vaccination than static models in cost-effectiveness analyses.

© 2023 Merck Sharp & Dohme LLC., a subsidiary of Merck & Co., Inc., Rahway, NJ, USA and The Author(s).

Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Contents

1. Introduction	5222
2. Methods	5222
2.1. Search strategy and databases	5222
2.2. Study selection	5222
2.3. Data extraction and synthesis	5223
2.4. Quality appraisal	5223
2.5. Risk of bias assessment	5223
3. Results	5223

* Corresponding author.

E-mail address: yao-hsuan.chen@msd.com (Y.-H. Chen).

3.1.	Key characteristics	5223
3.2.	Clinical and epidemiological parameters	5223
3.3.	Primary outcomes	5223
3.4.	Secondary outcomes	5223
3.5.	Costs	5225
3.6.	Utility weights	5225
3.7.	Quality appraisal	5225
3.8.	Risk of bias	5225
4.	Discussion	5227
5.	Conclusion	5231
	Declaration of Competing Interest	5231
	Acknowledgements	5231
Appendix A.	Supplementary material	5231
	References	5231

1. Introduction

Rotavirus is a leading cause of severe diarrhoea and dehydration in children under five years of age [1]. It is transmitted through the oral-faecal route, with nearly all children infected by the age of five years [1]. In low- and middle-income countries, rotavirus infections are associated with high levels of child mortality [2]. In high income countries, while rotavirus-related mortality is rare, the disease is a significant cause of morbidity and imposes a substantial economic burden on societies and health systems [3].

In 2009, Rotarix™ and RotaTeg™ were added to the World Health Organization (WHO) List of Essential Medicines [4]. However, rotavirus vaccination programmes have not been universally adopted, with 107 countries implementing vaccination programmes as of October 2020 [5]. As of 2016, only 45 % of high-income countries had introduced the vaccine [5]. Barriers to the implementation of universal rotavirus vaccination in high-income countries include poor knowledge among policy makers about the disease burden and its economic consequences, as well as a perception that rotavirus vaccination programmes are not cost-effective [6].

Previously published systematic reviews have presented evidence relating to the cost-effectiveness of rotavirus vaccines in different income settings [3,7–10]. In 2009, *Blicke and Beutels* reviewed 18 economic evaluations of rotavirus vaccination programmes, nine of which were in developed countries, in order to describe and assess methodological and modelling choices [7]. The review reported that “no unambiguous single answer emerged as to whether universal rotavirus vaccination is or is not cost-effective” [7]. The study concluded that future research should account for herd protection using dynamic transmission models. In 2013, *Aballéa et al.* described key differences in methods, assumptions and inputs, which were influential in driving differences in conclusions [8]. Like *Blicke and Beutels*, the review recommended the use of dynamic transmission models to account for herd protection in order to capture the full economic value of rotavirus vaccination. In 2016, *Thinboonboon et al.* assessed the quality of evidence from economic evaluations [9]. The study identified significant variation in quality between low- and middle-income and high-income countries, which was attributed, in part, to a lack of capacity among low- and middle-income countries to conduct cost-effectiveness analyses with local researchers. In 2017, *Koritum et al.* provided a global overview of evidence relating to the cost-effectiveness of rotavirus vaccines. The review included 104 studies, including 46 high-income countries, published between 2000 and 2015. 34 studies in high-income settings concluded rotavirus vaccination was cost-effective, whereas 12 concluded that rotavirus vaccination was not cost-effective [3]. In 2019, *Haider et al.* presented cost-effectiveness studies in low-income and lower-

middle-income countries, concluding that rotavirus vaccination is cost-effective in those settings [10]. These systematic reviews have included mainly static models, such as Markov models and decision trees, which do not account for herd immunity. However, herd immunity effects have been consistently demonstrated in large observational studies worldwide, particularly in high-income settings where large database studies are possible and primary data collection studies are frequent [11–15].

This systematic review aims to identify cost-effectiveness analyses that used dynamic transmission models in high-income countries. Results will be of interest to infectious disease modellers and policy makers interested in rotavirus vaccination.

2. Methods

This systematic review was undertaken according to the principles of systematic reviewing established in the Cochrane Handbook [16] and guidance document published by the Center for Reviews and Dissemination (CRD) of York University [17], United Kingdom. We used the PRISMA checklist for systematic reviews which is available in [Appendix I](#). The protocol was registered with PROSPERO (registration number: CRD42020208406).

2.1. Search strategy and databases

From 30 August 2020 – 3 September 2020, we systematically searched CEA Registry, MEDLINE, Embase, Health Technology Assessment Database, Scopus, and the National Health Service Economic Evaluation Database (NHS EED) for cost-effectiveness studies using dynamic transmission models in high-income countries. A full search strategy is provided in [Appendix II](#).

2.2. Study selection

Studies were eligible if they included: [1] children ≤ 5 years of age; [2] a high-income country; [3] a dynamic transmission model; [4] a live oral rotavirus vaccine; [5] incremental cost-effectiveness ratios (ICER) or incremental cost-utility ratios (ICUR).

Records were excluded if they were: [1] not a live, oral rotavirus vaccine; [2] not a cost-effectiveness analysis or cost-utility analysis; [3] not based on a dynamic transmission model; [4] not a high-income country; [5] not a full paper (conference papers, reviews and editorials were excluded); [6] published before 2002.

A country was considered as high-income if it was included on the World Bank list of high-income economies, which are defined based on their gross-national income (GNI) per capita [18]. A list of eligible countries is provided in [Appendix III](#).

Records written in languages other than English were eligible for inclusion. Two screeners independently translated these stud-

ies using Google Translate [19]. Full texts were screened, and discrepancies were resolved through discussion. Reasons for exclusion were recorded during full text screening.

The corresponding authors of each included study were contacted by email to identify further relevant studies. Additionally, we screened the abstracts of all titles included in the bibliographies of included studies, applying the same screening criteria to titles and abstracts as the initial screening. We also screened the titles and abstracts of studies that cited included studies, as well as all recommended readings associated with the included studies, which were provided by the journal. Further information is available in [Appendix IV](#).

2.3. Data extraction and synthesis

A data extraction form was created and piloted on two studies [20,21], available in [Appendix V](#). The data extraction form incorporated key characteristics of the two studies, including the setting, currency, price year used, analytical perspective, time horizon, main cost items, ICER/ICUR, and deaths averted. The data extraction form also included parameter values and sources for the dynamic transmission model specification and calibration.

2.4. Quality appraisal

All included studies were appraised using the Consensus on Health Economic Criteria (CHEC) list [22]. The CHEC list consists of 19 yes-or-no questions which highlight assumptions and limitations to ensure transparency when interpreting validity and generalisability.

Quality appraisal was conducted independently by two reviewers. After each reviewer completed the quality appraisal, results were compared, and any discrepancies were resolved through discussion. Any conflicts that could not be resolved were discussed with a third reviewer with the final decision made by a majority vote.

2.5. Risk of bias assessment

A risk of bias assessment was conducted for all included studies using the Philips checklist, which is recommended by the Cochrane Handbook [16]. The risk of bias assessment was conducted independently by one reviewer. The assessment was subsequently reviewed by a second reviewer. Feedback was provided and discussed with any discrepancies resolved through discussion. Any discrepancies that could not be resolved were discussed with a third reviewer.

3. Results

The search of electronic databases identified 3,900 papers across six databases ([Table 1](#)). All search results were exported to EndNote and duplicates were removed, after which 1,818 papers remained for title and abstract screening. In total, 61 references appeared to meet the inclusion criteria and moved to the full text screening stage. 57 records were excluded at the full text screening because either they were not a full study (three papers), the wrong study design on account of not being an economic evaluation (three papers), wrong study design on account of being a dynamic transmission model without an economic component (one paper), or wrong study design on account of being a static model (50 papers). Four papers met the inclusion criteria and were included in the review [20,21,23,24].

A screening of the bibliographies, citations or the recommended readings associated with the four papers identified no further stud-

ies for full text review or inclusion. Correspondence with authors identified no further studies for review.

3.1. Key characteristics

Four economic evaluations were identified from England and Wales, Norway, France, and the United States [20,21,23,24] ([Table 4](#)). Two models used variations of a susceptible-infected-recovered (SIR) structure [23,24], while a third study featured a susceptible-exposed-infected-recovered (SEIR) structure [21]. One study did not clearly describe its model structure [20]. Two studies explicitly modelled three total infections, and explicitly mentioned maternal immunity [21,24].

All four studies were cost-utility analyses, which used quality-adjusted life years (QALYs) as the primary measurement for health-related quality of life, and compared vaccination versus no vaccination [20,21,23,24]. One study was conducted from the healthcare perspective only [20]. Two studies used a healthcare and societal perspective [21,23] and one study used a third-party payer and societal perspective [24]. The time-horizon was five years [21], 20 years [23] and 50 years [20,24]. All studies applied discount rates, which varied between 3 and 4 %. Three studies used probabilistic sensitivity analyses [20,21,24], while one study was unclear about the methods used for sensitivity analysis [23].

3.2. Clinical and epidemiological parameters

The reporting of incidence data varied considerably. One paper reported incidence of severe rotavirus gastroenteritis (RVGE) as 43.6 cases per 1,000 children per year [20]. One paper described the probability, rather than incidence, of a child developing any RVGE as 0.8 [23]. The incidence of severe RVGE requiring hospitalisation was included in two papers, ranging from 3 per 1,000 [21] to 4.6 people per 1000 [24]. One paper presented the probability of requiring hospitalisation for severe RVGE as 0.0181 [23].

The duration of immunity following natural infection ranged from 18 months [24] to five years [23]. Duration of infection was presented as five days for severe and three days for mild infections in the United States [23]. All other studies reported the duration of infection as seven days for severe RVGE and 3.5 days for mild RVGE [20–21,24].

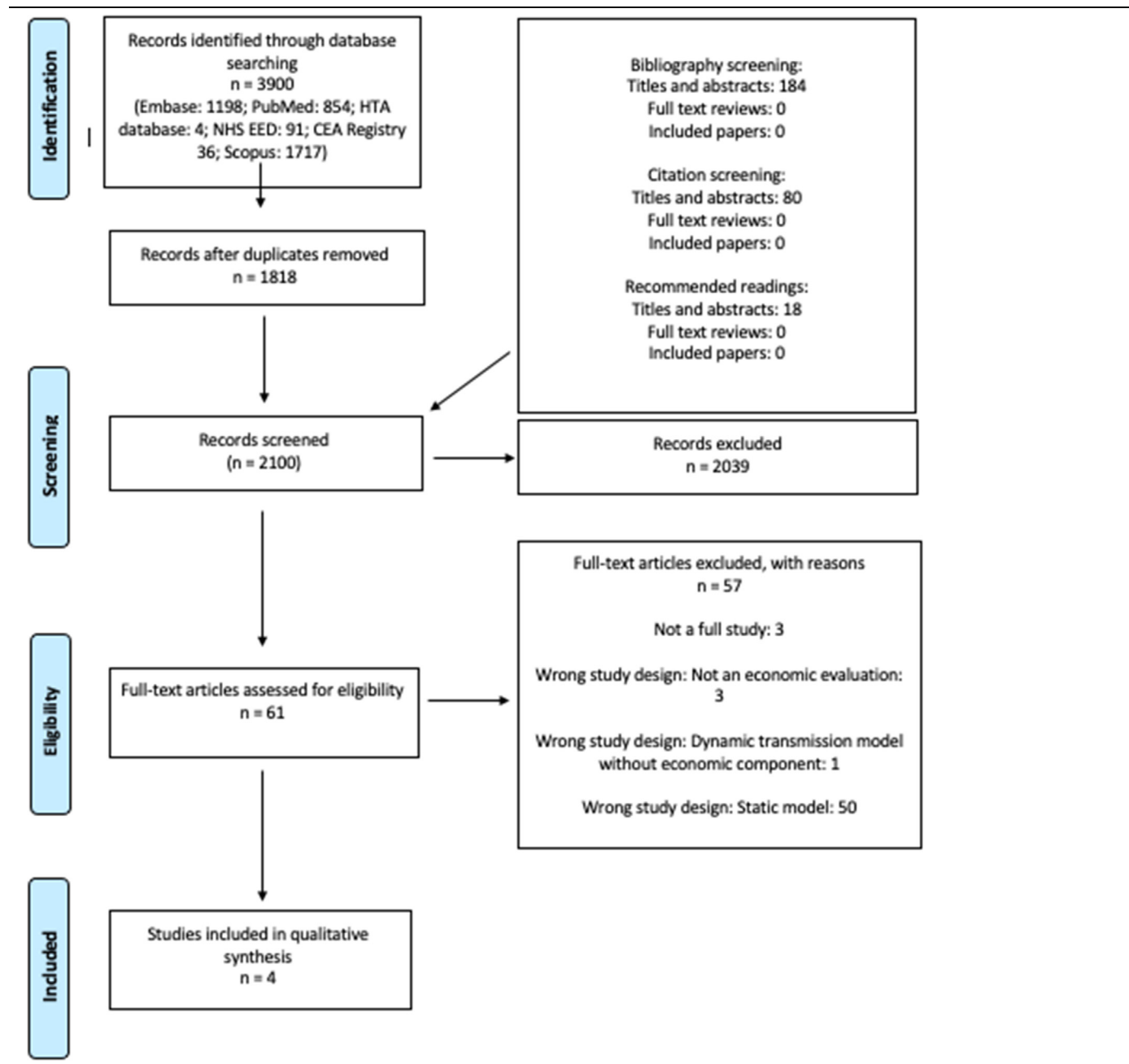
The proportion of severe RVGE was reported as 0.24 [20,23], while two studies modelled change in first and subsequent infections [21,24]. These studies reported a proportion of severe RVGE in primary infection of 0.13 [21,24], and proportions of severe RVGE in secondary infection of 0.03 [24] and 0.04 [21]. Further details regarding input parameters can be found in the [Appendix](#).

3.3. Primary outcomes

The cost per QALY gained ranged from cost-saving [20,21,23] to USD 192,100 [23]. Three studies found vaccination to be cost-effective from the healthcare perspective [20,21,23]. Two papers reported that vaccination would be cost-saving from a societal perspective [21,23]. One paper concluded it would be cost-saving from a healthcare perspective if it was administered concomitantly with DTaP/IPV/Hib vaccines and under the assumption that vaccine efficacy does not begin to wane immediately after administration [20].

Results were sensitive to vaccine price [20], vaccine immunity duration [20], vaccine administration costs [20], QALY loss of caregivers [24] and quality of life and workdays lost associated with homecare cases [21]. However, variation of the most sensitive parameters also concluded that vaccination was cost-effective.

Table 1
PRISMA Flow Diagram.



3.4. Secondary outcomes

All four studies reported that vaccination substantially decreased rotavirus-related outcomes including deaths, hospitalisations, general practitioner (GP) appointments, outpatient or emergency department visits, and cases.

Edwards *et al* estimated a 73 % reduction in deaths [21], while Shim *et al* estimated a 90 % reduction in deaths [23]. Shim *et al* reported an 89 % reduction in cases averted [23]. Yamin *et al* reported 185,507 cases averted at two years, 212,930 cases averted at five years, and 212,488 cases averted at ten years [24]. Edwards *et al* reported 156,610 cases averted (114,658 home cases; 34,571 primary care; 7,381 hospitalised) [21].

One study reported GP appointments averted, with 83,095 GP appointments averted at two years, 95,379 GP appointments averted at five years, and 95,629 GP appointments averted at 10 years [24].

Two studies reported outpatient or emergency department visits avoided [23,24]. Shim *et al* reported a 90 % reduction in outpatient or emergency department visits [23]. Yamin *et al* reported 3,536 outpatient or emergency department visits avoided at two years; 4,059 visits avoided at five years, and 4,069 visits avoided at ten years [24].

Three studies reported hospitalisations averted [21,23,24]. Shim *et al* reported a 90 % reduction in hospitalisations [23]. Yamin *et al* reported 12,498 hospitalisations averted at two years, 14,345

hospitalisations averted at five years, and 14,383 hospitalisations averted at ten years [24]. Edwards *et al.* reported 7,381 hospitalisations averted (73 %) [21].

3.5. Costs

Direct medical costs included the costs associated with health care resource utilisation (GP consultations, ED visits, nosocomial infections, inpatient stays and rotavirus death). Productivity losses were in three studies [21,23–24]. Three studies included administration time and costs [20–21,23].

3.6. Utility weights

Utility weights differed across all studies (Table 6). Utility decrements for severe RVGE or inpatient stays ranged from 0.00186 [23] to 0.00340 [21]. Utility decrements for mild RVGE or outpatient/primary care/home care cases ranged from 0.00055 [20] to 0.00186 [23]. Utility decrements for caregivers of severe RVGE/inpatients with RVGE was included in three studies [21,23,24] and ranged from 0.002 [23] to 0.0082 [19]. Utility decrements for caregivers of mild RVGE/inpatients with RVGE was included in three studies [21,23,24] and ranged from 0.00115 to 0.0046 [21].

Methods used to calculate utility weights in the primary data collection studies cited included the Health Utilities Index 2 for children [19], the EQ-5D-5L for parents of inpatients and outpatients [19] and the young children's quality of life questionnaire [23].

3.7. Quality appraisal

Table 2 presents the results of the quality appraisal. In total, Edwards *et al.* received 18 positive scores and one negative score,

Atkins *et al.* received 16 positive scores and three negative scores, Yamin *et al.* received 17 positive scores and two negative scores, and Shim *et al.* received 15 positive scores and four negative scores. A full description of the CHEC assessment for each study, including supporting comments, is available in Appendix VI.

Three patterns were identified, with two of the studies scoring negatively for question 17, which addresses the generalisability of the results to other settings and patient/client groups. Two studies scored negatively on question seven as they did not capture all relevant costs, including nosocomial infections which can influence cost-effectiveness results. Three studies scored negatively on question 18, which addresses whether studies indicate that there is no potential conflict of interest of study researcher(s) and funder(s).

3.8. Risk of bias

Table 3 shows that the risk of bias varied across studies. Edwards *et al.* sufficiently addressed 17 criteria, insufficiently addressed three criteria and provided insufficient information for assessment of two criteria. Atkins *et al.* sufficiently addressed 14 criteria, insufficiently addressed six criteria and provided insufficient information for assessment for two criteria. Yamin *et al.* sufficiently addressed 16 criteria, insufficiently addressed four criteria and provided insufficient information for assessment for two criteria. Shim *et al.* sufficiently addressed 10 criteria, insufficiently addressed five criteria and provided insufficient information for assessment for seven criteria. Patterns identified include all four studies failing to discuss the internal consistency of modelling choices, while two studies did not account for heterogeneity and two studies provided insufficient information regarding heterogeneity. Two studies insufficiently addressed structural or methodological uncertainty. A full justification of each score is provided in Appendix VII.

Table 2
Summary of results: Consensus Health Economic Criteria (CHEC list).

	Edwards <i>et al.</i>	Atkins <i>et al.</i>	Yamin <i>et al.</i>	Shim <i>et al.</i>
CHEC-list				
1 Is the study population clearly described?	+	+	+	+
2 Are competing alternatives clearly described?	+	+	+	+
3 Is a well-defined research question posed in answerable form?	+	+	+	+
4 Is the economic study design appropriate to the stated objective?	+	+	+	+
5 Is the chosen time horizon appropriate in order to include relevant costs and consequences?	+	+	+	+
6 Is the actual perspective chosen appropriate?	+	+	+	+
7 Are all important and relevant costs for each alternative identified?	–	+	+	–
8 Are all costs measured appropriately in physical units?	+	+	+	+
9 Are costs valued appropriately?	+	+	+	+
10 Are all important and relevant outcomes for each alternative identified?	+	+	+	–
11 Are all outcomes measured appropriately?	+	+	+	–
12 Are outcomes valued appropriately?	+	–	+	+
13 Is an incremental analysis of costs and outcomes of alternatives performed?	+	+	+	+
14 Are all future costs and outcomes discounted appropriately?	+	+	+	+
15 Are all important variables, whose values are uncertain, appropriately subjected to sensitivity analysis?	+	+	+	+
16 Do the conclusions follow from the data reported?	+	+	+	+
17 Does the study discuss the generalisability of the results to other settings and patient/client groups?	+	–	–	+
18 Does the article indicate that there is no potential conflict of interest of study researcher(s) and funder(s)?	+	–	–	–
19 Are ethical and distributional issues discussed appropriately?	+	+	+	+

Table 3
Summary of results for Philips risk of bias assessment.

		Edwards <i>et al.</i>	Atkins <i>et al.</i>	Yamin <i>et al.</i>	Shim <i>et al.</i>
Structure					
S1	Statement of decision problem/objective	+	+	+	+
S2	Statement of scope/perspective	+	+	+	+
S3	Rationale for structure	+	–	+	?
S4	Structural assumptions	+	+	+	+
S5	Strategies/comparators	+	–	–	+
S6	Model type	+	+	+	+
S7	Time horizon	–	+	+	?
S8	Disease states/pathways	+	+	+	+
S9	Cycle length	NA	NA	NA	NA
Data					
D1	Data identification	–	+	+	–
D2	Data modeling	+	?	+	?
D2a	Baseline data	+	–	+	+
D2b	Treatment effects	–	–	?	?
D2c	Costs	+	+	+	+
D2d	Quality of life weights (utilities)	+	+	+	+
D3	Data incorporation	+	–	+	–
D4	Assessment of Uncertainty	+	+	+	?
D4a	Methodological	+	+	–	–
D4b	Structural	+	+	–	–
D4c	Heterogeneity	?	–	–	?
D4d	Parameter	+	+	+	–
Consistency					
C1	Internal consistency	?	?	?	?
C2	External consistency	+	+	+	+

Table 4
Summary of key characteristics.

Study	Shim 2009	Atkins 2012	Yamin 2016	Edwards 2017
Location (setting)	USA	England and Wales	France	Norway
Target population	Children < 5 years	Children < 5 years	0–65+; cost and benefits focus on children younger than 5 years old	Birth Cohort (range 49 937–68 762 per year)
Model structure	Age-structured, states include susceptible, severely or mildly infected, temporarily recovered and vaccinated	Not specified in the methods.	Age-structured, Susceptible-Infected-Recovered (SIR) model including maternal immunity and up to 3 infections	Age-structured model with the following states: Maternal immunity, Susceptible, Exposed, Infected, Recovered, Susceptible, with reference to 3 and more infections
Intervention (comparator)	Vaccine vs no vaccination	Vaccine vs no vaccination	Vaccine vs no vaccination	Vaccine vs no vaccination
EE type	CUA	CUA	CUA	CUA
Currency unit and year	USD 2007	GBP 2011	EUR 2015	EUR 2015
Unit of benefit	QALY	QALY	QALY	QALY
Perspective	Healthcare and societal	Healthcare	Third-party payer (TPP) and societal	Healthcare and societal
Time horizon	20 years	50 years	50 years	2015–2019
Discount rate	3 %	3.5 % (0–6 %)	4 % annually for the first 30 years, decreasing linearly to 2 % annually by 50 years	4 %
Sensitivity analysis	A sensitivity analysis was conducted by varying key parameters. The method was not clearly described	Probabilistic sensitivity analysis	Univariate sensitivity analysis; probabilistic global uncertainty analysis, Monte Carlo simulations	One-way, multi-way, and probabilistic sensitivity analysis with Monte Carlo simulations
Study funder	Not reported	Sanofi-Pasteur MSD; Miriam Weston Trust	Not reported	This study received no funding

Note: EE = Economic Evaluation; CUA = Cost-Utility Analysis; QALY = Quality Adjusted Life Years.

Table 5
Summary of results.

Study	Shim 2009	Atkins 2012	Yamin 2016	Edwards 2017
Conclusion	Vaccination was cost effective from healthcare perspective when we considered child with one caregiver; cost-saving from societal perspective	Vaccination was cost effective, possibly cost-saving	Vaccination was cost effective from societal and third payer perspectives	Vaccination was cost effective from healthcare perspective; cost-saving from societal perspective.
Primary outcomes ICER/ICUR	The cost per QALY gained was found to be USD 148,270–192,100 per child only and USD 80,740–104,610 when the authors considered a child with one caregiver.	GBP 45/course, GBP/QALY: (A) 12,902 (immediate waning concomitant administration); (B) –3,222 (delayed waning concomitant administration); (C) 32,826 (immediate waning separate administration); (D) 10,517 (delayed waning separate administration) GBP 60/course, GBP/QALY: (A) 27,133 (immediate waning concomitant administration); (B) 6,563 (delayed waning concomitant administration); (C) 47,057 (immediate waning separate administration); (D) 20,402 (delayed waning separate administration)	Societal perspective: EUR 28,500 and EUR 39,500 per QALY gained using vaccine prices of EUR 115 and EUR 135. From the third-party payer perspective, incremental cost-effectiveness ratios ranged from EUR 12,500 to EUR 20,000/QALY	From the healthcare perspective, vaccination ICER was EUR 47,447 – 52,709 per QALY
Secondary outcomes				
Deaths averted	90 % reduction	NA	Average annual deaths averted at: 2 years: 8 5 years: 9 10 years: 9	Mean 1.996 (73 % reduction)
Cases averted	89 % reduction	NA	Average cases averted at: 2 years: 185,507 5 years: 212,930 10 years: 212,488	156,610 (114,658 home cases; 34,571 primary care; 7,381 hospitalized)
GP appointments averted	NA	NA	2 years: 83,095 5 years: 95,379 10 years: 95,629	NA
Outpatient or emergency department visits avoided	90 % reduction	NA	2 years: 3,536 5 years: 4,059 10 years: 4,069	NA
Hospitalisations averted	90 % reduction	NA	2 years: 12,498 5 years: 14,345 10 years: 14,383	7,381 (73 % reduction)
QALYs gained / DALYs averted	NA	With vaccination 7,800 QALYs lost; without vaccination 24,500 QALYs lost	Dynamic model: 14,163 Static model: 18,153	70 % decrease in QALYs lost due to rotavirus
Herd immunity effect	NA	Accounting for herd protection reduces the threshold price of cost-effectiveness by between 4 % and 13 % for concomitant vaccine administration	Compared with the static model, the dynamic model reduces the ICUR for the TPP by EUR 3,500 and EUR 4,000 for a EUR 114 per vaccine course and EUR 135 per vaccine course respectively.	NA

(continued on next page)

Table 5 (continued)

Study	Shim 2009	Atkins 2012	Yamin 2016	Edwards 2017
			From the societal perspective, compared with the static model, the dynamic model reduces the ICUR for the TPP EUR 5,500 and EUR 7,000 for a EUR 114 per vaccine course and EUR 135 per vaccine course respectively	
Change in hospitalisation rates of children with rotavirus infection	NA	NA	Average cases at 2 y: 12,490 Average cases at 5 y: 14,345 Average cases at 10 y: 14,383	Decrease of 73 %
Change in primary cases and home cases rates of children with rotavirus infection	NA	NA	GP visits Average cases at 2 y: 83,095 Average cases at 5 y: 95,379 Average cases at 10 y: 95,629	Primary cases reduced by 70 % and home cases reduced by 64 %
Change in incidence of rotavirus infection	NA	Incidence no vaccination Severe RVGE per million: 279 Mild RVGE per million: 882 Incidence with vaccination Severe RVGE per million: 32.4 Mild RVGE per million: 307	The incidence reduction in children younger than 5 years ranges from 62 % to 65 % (2yearsaftervaccineintroduction) to 71 % to 75 % (10yearsaftervaccineintroduction)	NA
<i>Other</i>				
Cost-effectiveness threshold used	3x GDP/capita	GBP 20,000–30,000	EUR 30,000 and EUR 50,000	EUR 73,444
Most sensitive parameter	NA	Vaccine price; vaccine immunity duration; vaccine administration cost	QALY losses of caregivers	The health-related quality of life and workdays lost associated with homecare cases

Table 6
Summary of costs and utilities.

Study	Shim 2009	Atkins 2012	Yamin 2016	Edwards 2017
Healthcare costs accounted for	Direct medical costs, including vaccination, administration and adverse events for the following scenarios: death, hospitalisation, ED, hospital outpatient visit, office visit	Calls to NHS Direct, GP consultations, A&E consultations, Hospital admissions	GP consultations, ED visits, Hospitalisations/deaths, Nosocomial infections	The total cost of the vaccination programme, including the vaccine itself, and also the implementation, administration and storage costs
Non-healthcare direct costs (e.g. patient/family) accounted for	Travel costs, extra diapers, and childcare costs	NA	Travel costs, extra diapers, childcare	NA
Societal costs (e.g. productivity) accounted for	Foregone earning of parents, lifetime productivity loss of a dying child	NA	NA	Productivity losses due to work absences
<i>Unit costs</i>				
GP consultation	USD 120	GBP 36 Sensitivity analysis: 32.40–39.60	Third party payer: EUR 36.64 Societal perspective: EUR 67.5	EUR 24
Outpatient visit (tertiary care)	USD 488	NA	NA	EUR 214
Emergency department (ED) consultation	USD 551	GBP 111 Sensitivity analysis: 99.90–122.10	TPP: EUR 86.2 SP: EUR 142.2	EUR 38
Inpatient stay	USD 3,958	GBP 1,047.96 Sensitivity analysis: 943.16–1,152.76	Hospitalisations/deaths: TPP: EUR 1,331.5 SP: EUR 1,385.5	NA
Daily productivity loss of caregiver (unit value)	Homecare episodes: Foregone earnings of parent (for 1 day): USD 134 Healthcare episodes except death Foregone earnings of parent (for 1.3 days): USD 174 Vaccination Foregone earnings of parent (for 1 day in 25 % of cases): USD 34 Vaccine adverse effects: Foregone earnings of parent (for 1 day): USD 134 Death: Foregone earnings of parent (for 10 days): USD 1,340	NA	EUR 46 per episode of RVGE not seeking medical treatment For GP visit EUR 239.547 per episode For ER visit EUR 11.07 per visit EUR 270 per hospitalisation episode	EUR 347
Days of productivity lost in GP consultation and ED	1.3 days	NA	NA	1.5
Days of productivity lost in- and outpatients	1.3 days	NA	NA	3.25
Days of productivity lost in homecare cases	1 day	NA	2.5 h loss of work per episode (0.10 days)	1
<i>Vaccination cost</i>				
Baseline cost of fully vaccinated child (number of doses, price per dose)	USD 245 - USD 225	In the base case scenario, values of GBP 45 and GBP 60 per full-course regimen were used	In the base case scenario, values of EUR 115 and EUR 135 per full-course regimen were used	EUR 54
Administration time of health care workers	USD 10	NA	NA	20.5 to 25 min
Health worker cost for administration	NA	Cost of vaccine course admin, concomitant: GBP 15 Cost of vaccine course admin, separate: GBP 36	NA	EUR 424.5 per hour

(continued on next page)

Table 6 (continued)

Study	Shim 2009	Atkins 2012	Yamin 2016	Edwards 2017
Caregiver productivity loss for administration	1 day in 25 % of cases	NA	NA	NA
Implementation and operational costs of vaccination programme (vaccine, implementation, administration, and storage costs)	NA	NA	NA	EUR 335,083 per year
Utility weights				
Quality of life lost/utility decrement for severe RVGE or inpatient stay	0.00186	0.0022	0.003274	0.00340
Quality of life lost/utility decrement for mild RVGE or outpatient/primary care/home care cases	0.00186	Sensitivity analysis: (0.00055, 0.0000165) (0.00055)	0.001715	Home and primary care: 0.00115
Caregivers of severe RVGE/inpatients with RVGE	0.002	NA	0.003274	Inpatient caregivers: 0.0082
Caregivers of mild RVGE/outpatients with RVGE	0.002	NA	0.001715	Outpatient caregivers: 0.0046

Note: A&E: Accident & Emergency; ED = Emergency Department; ER = Emergency Room; GP = General Practitioner; RVGE = Rotavirus Gastroenteritis; SP = Societal Perspective; TPP = Third Party Payer.

4. Discussion

This review identified four economic analyses that used dynamic transmission modelling to evaluate the cost-effectiveness of rotavirus vaccination. Consistent with results of dynamic transmission models in previous systematic reviews, all included studies concluded that rotavirus vaccination was cost-effective in high-income settings from both societal and healthcare perspectives when compared with no vaccination (see Table 5).

The impact of vaccine herd-protection effects in cost-effectiveness analyses of childhood vaccination has been noted previously. Two systematic reviews on the cost-effectiveness of rotavirus vaccination have recommended that future economic evaluations use dynamic transmission models to account for herd immunity [7,8]; while a quantitative comparative analysis of childhood vaccinations published in 2017 concluded that “when the target-vaccination strategy was not cost-saving without herd-protection, inclusion of herd-protection always resulted in more favourable results” [25]. Dynamic transmission modelling was also recommended in the WHO guide for standardization of economic evaluations of immunization programmes for infectious diseases that involve human to human transmission and where one of the eligible target groups includes an “epidemiologically influential subgroup”, such as children [26]. Still, as highlighted by this review, few economic evaluations have featured dynamic transmission modelling to determine the cost-effectiveness of rotavirus vaccination programmes.

Static models that ignore herd immunity effects can underestimate the cost-effectiveness of vaccination and introduce structural bias against vaccinations for infectious diseases. Brisson *et al.* demonstrated that structural uncertainty, including modelling choices, could have equal or greater impacts on results than parameter uncertainty [27]. Of note, these assumptions and modelling choices are also rarely included in sensitivity analyses. Kori-tum *et al.* analysed eight cost-utility studies using static models, which compared universal rotavirus vaccination to no intervention in the United Kingdom, France, and the USA [3]. Out of the eight studies, six studies concluded that rotavirus vaccination was not cost-effective. Similarly, a report in 2009 featuring an economic evaluation using a static model in Norway concluded that rotavirus vaccination would not be cost-effective [28].

Herd immunity effects after administration of rotavirus vaccine have been observed in the United States, Australia, Finland, and several other high-income countries where database studies are possible [11–15]. Following introduction of rotavirus vaccination, decreases in rotavirus incidence have been observed in children, adults, and the elderly. This is consistent with household transmission studies that demonstrated that rotavirus was highly transmissible within households, thus suggesting that averting cases in children potentially averts cases in older siblings and relatives [29–31]. Of note, post-licensure analyses and modelling of rotavirus vaccination in the United States, Australia, and Finland have suggested vaccination to be cost-saving, with Finnish post-licensure observational data indicating that for every fully vaccinated child there were 33 Euros (2015E) in savings from a health care payer perspective [32,33].

Economic analyses that do not feature herd immunity effects thus risk underestimating the full value of vaccination. Analyses for other diseases have dealt with this challenge by considering herd-immunity effects directly in Markov models [34]. Of note, a limitation of the current study is that it only looked at high-income settings. Post-licensure studies documenting herd immunity effects in low- and middle-income settings are scarce. Further studies are required to ascertain the magnitude of herd immunity effects in low- and middle-income countries, where rotavirus vaccines have been shown to be less effective.

The studies included varied significantly in their methodological choices and included parameters. Notably, included studies utilised different model types, including M–SEIR, SIR–V, M–SIR, while one study did not clearly define its model structure. Variation between model types could also result in structural biases that are likely to influence the results of each study. Increased standardisation and guidance for economic modelling would facilitate better comparisons across studies. Still, across all reviewed studies, rotavirus vaccination was consistently identified as a cost-effective intervention from both societal and health system perspectives, indicating the results are robust to variation in parameter and structural decisions within a dynamic transmission model framework.

Similarly, variation was identified in sensitivity analyses, with studies employing different methods, including one multi way, multi-way, and probabilistic sensitivity analysis with Monte Carlo simulations. Across the included studies, the most sensitive parameter varied between vaccine price, vaccine immunity waning, health-related quality of life and workdays lost associated with homecare cases and QALY consideration for caregivers. It is not known to what extent methodological and parameter choices influenced the results of each study.

5. Conclusion

This study identified four economic evaluations that concluded rotavirus vaccination was cost-effective from both societal and health system perspectives. These studies allowed for rotavirus vaccination to impact disease transmission and thus unvaccinated individuals, consistent with large observational studies in high-income countries. As shown in other systematic reviews, only a minority of models have taken herd immunity into account, possibly underestimating the cost-effectiveness of rotavirus vaccination in a way that cannot be captured via sensitivity analyses. Thus, dynamic transmission models remain the most adequate models to quantify the benefits of rotavirus vaccination in high-income countries, with static models providing only conservative estimates of the value of rotavirus vaccination.

Data availability

All data used in this review are publicly available.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This project was developed with financial support from Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA via a project implemented by Triangulate Health Ltd.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.vaccine.2023.06.064>.

References

- [1] Rotavirus vaccines WHO position paper: January 2013 - Recommendations. Vaccine [Internet]. 2013; 31(52): 6170–1. Available from: <https://www.who.int/wer/2013/wer8805.pdf?ua=1>
- [2] Troeger C, Khalil IA, Rao PC, Cao S, Blacker BF, Ahmed T, et al. Rotavirus vaccination and the global burden of rotavirus Diarrhea among children younger than 5 years. JAMA Pediatr 2018;172(10):958–65.
- [3] Kotirum S, Vutipongsatorn N, Kongpakwattana K, Hutubessy R, Chaiyakunapruk N. Global economic evaluations of rotavirus vaccines: A systematic review. Vaccine [Internet]. 2017;35(26):3364–86. , <https://login.proxy.library.uu.nl/login?url=http://ovidsp.ovid.com/ovidweb.cgi?T=JS>.
- [4] World Health Organization. Rotavirus: immunization, vaccines and biologicals [Internet]. 2020 [cited 2020 Oct 20]. Available from: <https://www.who.int/immunization/diseases/rotavirus/en/>.
- [5] The Rota Council. Rota: global introduction status [Internet]. [cited 2020 Oct 20]. Available from: <http://preventrotavirus.org/vaccine-introduction/global-introduction-status/>.
- [6] Parez N, Giaquinto C, Du Roure C, Martinon-Torres F, Spoulou V, Van Damme P, et al. Rotavirus vaccination in Europe: drivers and barriers. Lancet Infect Dis 2014;14(5):416–25.
- [7] Bilcke J, Beutels P. Reviewing the cost effectiveness of rotavirus vaccination: The importance of uncertainty in the choice of data sources. Pharmacoeconomics 2009;27(4):281–97.
- [8] Aballea S, Millier A, Quilici S, Carroll S, Petrou S, Toumi M. A critical literature review of health economic evaluations of rotavirus vaccination. Hum Vaccines Immunother 2013;9(6):1272–88.
- [9] Thinboonboon K, Santatiwongchai B, Chantaratapornchit V, Rattanavipapong W, Teerawattananon Y. A systematic review of economic evaluation methodologies between resource-limited and resource-rich countries: a case of rotavirus vaccines. Appl Health Econ Health Policy 2016;14(6):659–72.
- [10] Haider S, Chaikledkaew U, Thavorncharoensap M, Youngkong S, Islam MA, Thakkestian A. Systematic review and meta-analysis of cost-effectiveness of rotavirus vaccine in low-income and lower-middle-income countries. Open Forum Infect Dis [Internet] 2019;6(4).
- [11] Paule-Korinek et al. Herd immunity after two years of the universal mass vaccination program against rotavirus gastroenteritis in Austria. Vaccine 2011;29(15). <https://www.sciencedirect.com/science/article/pii/S0264410X11001769>.
- [12] Mast T Christopher, Wang Florence T, Su Sue, Seeger John D. Evidence of herd immunity and sustained impact of rotavirus vaccination on the reduction of rotavirus-related medical encounters among infants from 2006 through 2011 in the United States. Pediatr Infect Dis J 2015;34(6):615–20. <https://doi.org/10.1097/INF.0000000000000702>.
- [13] Tate J et al. Sustained decline in rotavirus detections in the United States following the introduction of rotavirus vaccine in 2006. Pediatr Infect Dis J 2011;30(1). <https://doi.org/10.1097/INF.0b013e3181ffe3eb>.
- [14] Wilson, et al. Population-level impact of Ontario's infant rotavirus immunization program: evidence of direct and indirect effects. PLOS ONE 2016. Available at: <https://doi.org/10.1371/journal.pone.0154340>.
- [15] Yandle, et al. Indirect impact of rotavirus vaccination on viral causes of acute gastroenteritis in the elderly. J Clin Virol 2021; 137. Available at: <https://doi.org/10.1016/j.jcv.2021.104780>.
- [16] Cochrane. Cochrane Handbook for Systematic Reviews of Interventions [Internet]. 2020 [cited 2020 Sep 1]. Available from: <https://training.cochrane.org/handbook/current>.
- [17] Centre for Reviews and Dissemination guidance for undertaking reviews in health care [Internet]. Centre for Reviews and Dissemination. [cited 2020 Sep 4]. Available from: https://www.york.ac.uk/media/crd/Systematic_Reviews.pdf.
- [18] World Bank Country and Lending Groups [Internet]. World Bank; 2020 [cited 2020 Sep 2]. Available from: <https://datahelpdesk.worldbank.org/knowledgebase/articles/906519-world-bank-country-and-lending-groups>
- [19] Google Translate [Internet]. Google (Alphabet inc); 2020 [cited 2020 Sep 16]. Available from: <https://translate.google.com/>.
- [20] Atkins KE, Shim E, Carroll S, Quilici S, Galvani AP. The cost-effectiveness of pentavalent rotavirus vaccination in England and Wales. Vaccine [Internet] 2012;30(48):6766–76. , <https://login.proxy.library.uu.nl/login?url=http://ovidsp.ovid.com/ovidweb.cgi?T=JS&CSC=Y&NEWS=N&PAGE=fulltext&D=med8&AN=23000223>.
- [21] Hansen Edwards C, De Blasio BF, Salaman BV, Flem E. Re-evaluation of the cost-effectiveness and effects of childhood rotavirus vaccination in Norway. PLoS One 2017;12(8):1–19.
- [22] Evers S, Goossens M, de Vet H, van Tulder MAA. Criteria list for assessment of methodological quality of economic evaluations: Consensus on Health Economic Criteria. Int J Technol Assess Heal Care 2005;21(2):240–5.
- [23] Shim E, Galvani AP. Impact of transmission dynamics on the cost-effectiveness of rotavirus vaccination. Vaccine [Internet] 2009;27(30):4025–30. , <http://www.crd.york.ac.uk/CRDWeb/ShowRecord.asp?ID=22009102048>.
- [24] Yamin D, Atkins KE, Remy V, Galvani AP. Cost-effectiveness of rotavirus vaccination in France—accounting for indirect protection. Value Heal [Internet] 2016; 19(6): 811–9. <http://dx.doi.org/10.1016/j.jval.2016.05.011>.
- [25] Holubar M, Stavroulakis MC, Maldonado Y, Ioannidis JPA, Contopoulos-Ioannidis D. Impact of vaccine herd-protection effects in cost-effectiveness analyses of childhood vaccinations. A quantitative comparative analysis. PLoS One [Internet] 2017;12(3). , <https://www.scopus.com/inward/record.uri?eid=2-s2.0-85014361362&doi=10.1371%2Fjournal.pone.0172414&partnerID=40&md5=39318ce7a92686736d06c6e8d5397005>.
- [26] World Health Organization. WHO Guide on Standardization of Economic Evaluations of Immunization Programmes; 2019 (October). Available from: https://www.who.int/immunization/documents/who_ivb_19.10/en/

- [27] Brisson M, Edmunds WJ. Impact of model, methodological, and parameter uncertainty in the economic analysis of vaccination programs. *Med Decision Making* [Internet] 2006;26(5):434–46.
- [28] Bakgrunn M. Kostnadseffektivitet av å inkludere vaksinasjon mot rotavirus i det norske barnevaksinasjons- programmet; 2009.
- [29] Quee FA, de Hoog MLA, Schuurman R, Bruijning-Verhagen P. Community burden and transmission of acute gastroenteritis caused by norovirus and rotavirus in the Netherlands (RotaFam): a prospective household-based cohort study. *Lancet Infect Dis* 2020;20(5):598–606. [https://doi.org/10.1016/S1473-3099\(20\)30058-X](https://doi.org/10.1016/S1473-3099(20)30058-X). Epub 2020 Feb 20 PMID: 32087775.
- [30] Wikswo ME, Parashar UD, Lopman B, Selvarangan R, Harrison CJ, Azimi PH, et al. Evidence for household transmission of rotavirus in the United States, 2011–2016. *J Pediatric Infect Dis Soc* 2020;9(2):181–7. <https://doi.org/10.1093/jpids/piz004>. PMID: 30753568.
- [31] Lopman B, Vicuña Y, Salazar F, Broncano N, Esona MD, Sandoval C, et al. Household transmission of rotavirus in a community with rotavirus vaccination in Quinde, Ecuador. *Ecuador. PLoS One* 2013;8(7):e67763.
- [32] Newall AT, Leong RN, Reyes JF, Curns AT, Rudd J, Tate J, et al. Rotavirus vaccination likely to be cost saving to society in the United States. *Clin Infect Dis* 2021;73(8):1424–30. <https://doi.org/10.1093/cid/ciab442>. PMID: 34038527.
- [33] Reyes JF, Wood JG, Beutels P, Macartney K, McIntyre P, Menzies R, et al. Beyond expectations: Post-implementation data shows rotavirus vaccination is likely cost-saving in Australia. *Vaccine* 2017;35(2):345–52. <https://doi.org/10.1016/j.vaccine.2016.11.056>. PMID: 27916411.
- [34] Stoecker C, Hampton LM, Link-Gelles R, Messonnier ML, Zhou F, Moore MR. Cost-effectiveness of using 2 vs 3 primary doses of 13-valent pneumococcal conjugate vaccine. *Pediatrics* 2013;132(2):e324–32. <https://doi.org/10.1542/peds.2012-3350>.